TRIUNE BIBLICAL UNIVERSITY

GLOBAL EXTENSION · USA

BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Introduction to Artificial Intelligence

AI.101 · ORIENTATION LECTURE

A guided foundation in intelligent agents, AI methods, learning systems, robotics, and responsible practice.

NINE-HOUR LEARNING JOURNEY

Welcome to AI.101

HOUR 01

- Bachelor of Artificial Intelligence and Robotics Engineering
- Course: Introduction to Artificial Intelligence · 2.70 credits
- A foundation for thinking clearly about intelligent systems

KEY IDEA

Begin with questions, evidence, and responsible curiosity.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

The promise of this course

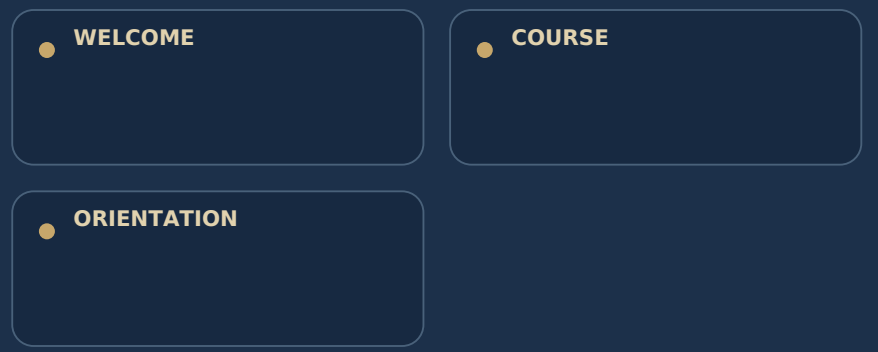
HOUR 01

- Build a usable vocabulary for AI
- Recognize major AI approaches and where they fit
- Connect data, models, decisions, and physical action

KEY IDEA

AI is a family of methods—not one magic machine.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Learning outcomes

HOUR 01

- Explain core AI concepts in your own words
- Trace a simple AI system from input to output
- Compare rule-based and learning-based approaches
- Identify risks, limitations, and human responsibilities

KEY IDEA

Understanding comes before implementation.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Where AI.101 sits in the degree

HOUR 01

- Year 1: foundations in AI, robotics, learning, vision, language, and control
- Year 2: perception, autonomous systems, deep learning, and kinematics
- Years 3–4: integration, learning, manipulation, ethics, and capstone work

KEY IDEA

This course gives later technical courses a shared foundation.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

A map of the discipline

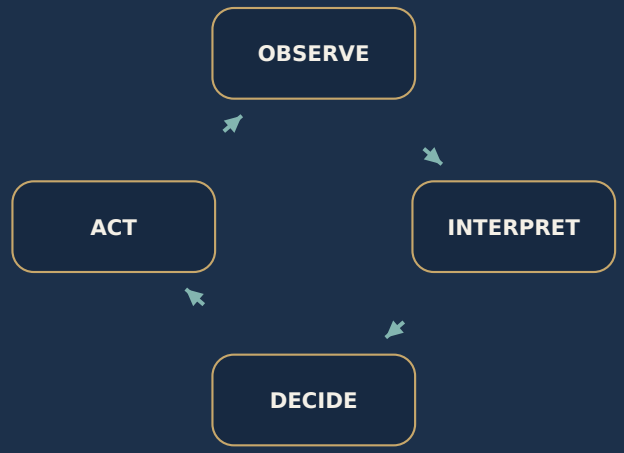
HOUR 01

- Perception: interpret signals from the world
- Reasoning: represent facts and draw conclusions
- Learning: improve behavior from experience or data
- Action: choose and execute a response

KEY IDEA

A complete system links sensing, thinking, and doing.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

How to learn from this presentation

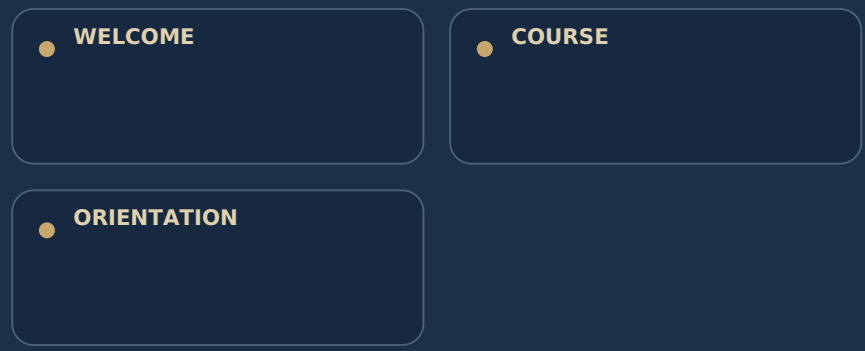
HOUR 01

- Pause at each “think” prompt before revealing the answer
- Sketch diagrams by hand; explain them aloud
- Keep a running glossary of unfamiliar terms
- Treat examples as starting points for inquiry

KEY IDEA

Active recall makes concepts more durable.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Nine-hour learning rhythm

HOUR 01

- Nine guided learning blocks, one per hour
- Each block mixes explanation, visual walkthrough, reflection, and knowledge check
- Pause between blocks; use the activities rather than watching passively

KEY IDEA

The schedule is designed for paced orientation—not nine hours of uninterrupted talking.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Course norms: rigor and humility

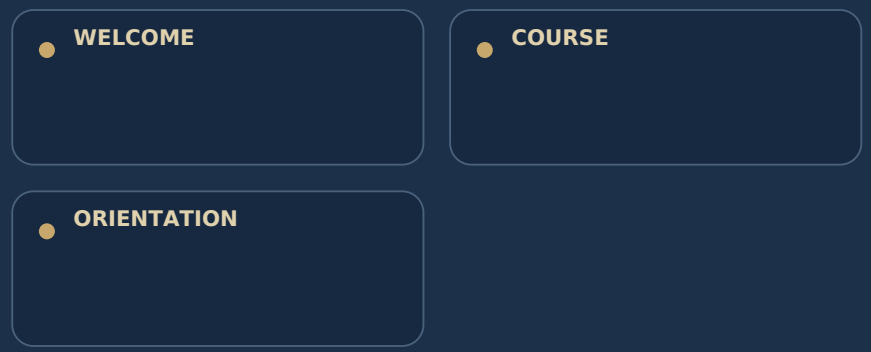
HOUR 01

- Ask what evidence supports a claim
- Distinguish a model’s output from ground truth
- Name uncertainty rather than hiding it
- Respect people affected by automated decisions

KEY IDEA

Good engineering combines capability with accountability.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Opening reflection: where do you meet AI?

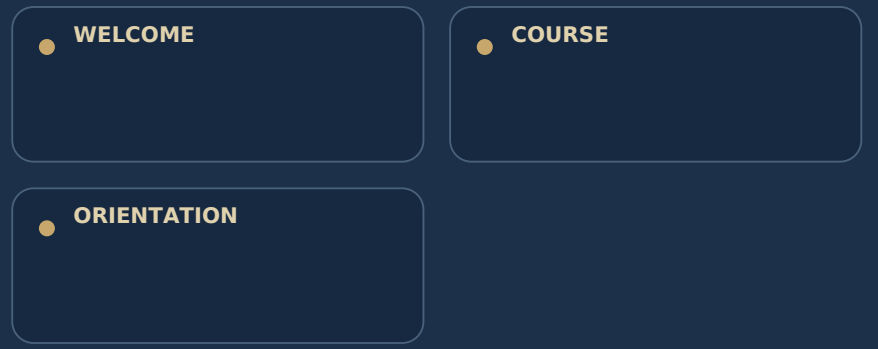
HOUR 01

- Name one system you use that may contain AI
- What does it take as input?
- What does it produce or influence?
- What could go wrong if its output is wrong?

KEY IDEA

Start with the system’s role and consequences.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 1 synthesis

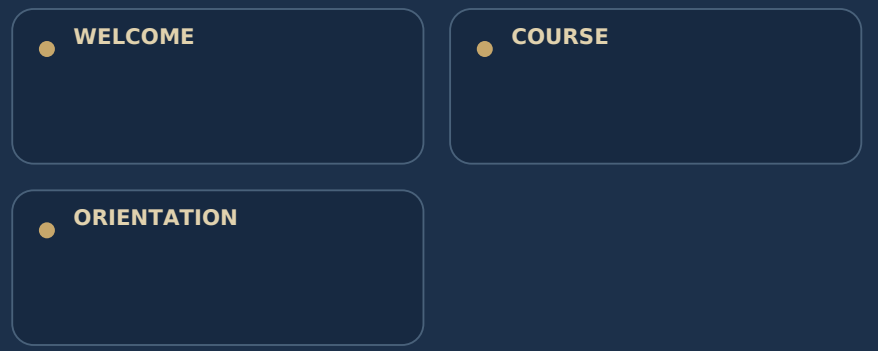
HOUR 01

- AI is a broad field with many approaches
- The degree connects algorithms to embodied machines
- Clear questions and disciplined evidence are our starting tools

KEY IDEA

Next: define artificial intelligence carefully.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

A working definition

HOUR 02

- AI studies computational systems that perform tasks associated with intelligent behavior
- Tasks include perception, prediction, language, planning, and decision support
- Definitions evolve as capabilities and expectations change

KEY IDEA

Define the task and context before labeling a system “intelligent.”

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Intelligence is not one score

HOUR 02

- Different tasks require different capabilities
- Competence in one setting does not imply general competence
- Human intelligence includes social, physical, and contextual dimensions

KEY IDEA

Measure the capability that matters for the application.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Agent and environment

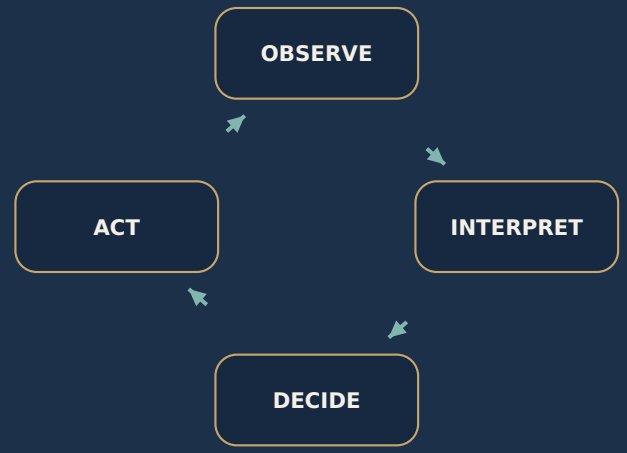
HOUR 02

- An agent receives observations through sensors or data inputs
- It selects actions through an actuator, interface, or recommendation
- The environment changes in response, producing new observations

KEY IDEA

An agent is defined by its interaction loop.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

A simple agent loop

HOUR 02

- Observe: collect the current state or signal
- Interpret: estimate what is happening
- Choose: select an action using rules or a model
- Act and evaluate: observe what changed

KEY IDEA

Feedback closes the loop.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Task environments

HOUR 02

- Fully or partially observable
- Deterministic or stochastic
- Episodic or sequential
- Static or changing; single-agent or multi-agent

KEY IDEA

Environment properties shape which methods are suitable.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

AI, automation, and software

HOUR 02

- Automation follows prescribed procedures for repeatable tasks
- AI methods infer, search, learn, or reason under variation
- Many products combine conventional software with AI components

KEY IDEA

Not every automated feature is AI.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Narrow capability and generality

HOUR 02

- Most deployed AI is designed for bounded tasks
- A system can be highly capable and still fail outside its design conditions
- Broad general intelligence remains a contested research concept

KEY IDEA

Avoid inferring general understanding from narrow performance.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Case: a campus help assistant

HOUR 02

- Inputs: student question and approved knowledge sources
- Process: retrieve relevant material and compose a response
- Output: answer that may guide a student decision
- Safeguards: source links, escalation, and correction path

KEY IDEA

Useful answers need provenance and a human route for unresolved cases.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Think: is it AI?

HOUR 02

- A thermostat switching at a fixed threshold
- A camera classifying plant disease from images
- A spreadsheet calculating a known formula
- A robot adapting its route from sensor readings

KEY IDEA

Discuss: what makes each case fixed, inferred, or adaptive?

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 2 synthesis

HOUR 02

- AI is best understood through tasks, methods, and operating conditions
- Agents connect observation to action
- The environment determines what success and failure mean

KEY IDEA

Next: follow AI’s history through changing ideas and tools.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Why study AI history?

HOUR 03

- Ideas return in new technical forms
- Progress depends on data, compute, algorithms, and expectations
- Historical cycles teach caution about exaggerated promises

KEY IDEA

History helps separate durable ideas from temporary hype.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Early questions and formal roots

HOUR 03

- Can reasoning be represented as symbol manipulation?
- Can computation solve problems through search?
- How should a machine’s behavior be evaluated?
- Logic, mathematics, and early computing shaped the field

KEY IDEA

AI grew from several disciplines, not a single invention.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Symbolic AI and search

HOUR 03

- Represent states, actions, and goals explicitly
- Explore candidate paths through a search space
- Use heuristics to prioritize promising possibilities

KEY IDEA

Explicit representations can be interpretable, but brittle when the world is messy.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Knowledge-based systems

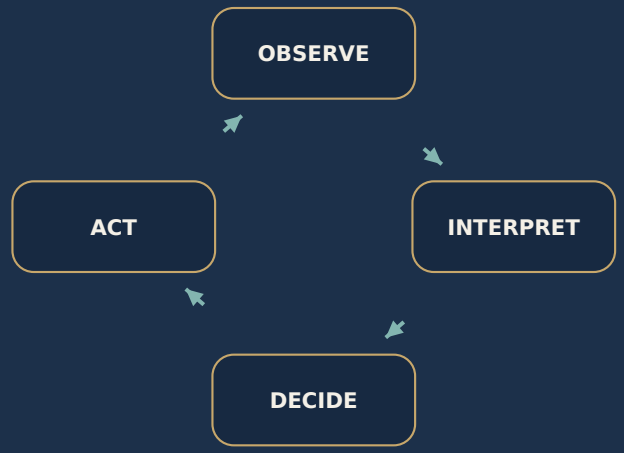
HOUR 03

- Expert knowledge encoded as facts and rules
- Inference engines apply rules to reach conclusions
- Useful in constrained domains with accessible expertise

KEY IDEA

Knowledge acquisition and maintenance became major bottlenecks.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Learning from data

HOUR 03

- Statistical methods estimate patterns from examples
- Features translate raw data into useful measurements
- Evaluation tests whether patterns generalize beyond training cases

KEY IDEA

The shift made data quality and evaluation central.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Neural networks and deep learning

HOUR 03

- Layered networks learn representations from data
- More data and compute enabled larger models
- Deep learning advanced speech, vision, and language tasks

KEY IDEA

Scale can improve performance, but does not remove uncertainty.

VISUAL MODEL



signals become relationships

Conceptual illustration · not to scale

Modern generative models

HOUR 03

- Models learn statistical structure from large datasets
- They generate plausible sequences, images, audio, or code
- Plausibility is not the same as factual correctness

KEY IDEA

Generated output must be checked against trusted evidence.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Recurring AI winters and hype

HOUR 03

- Expectations can outrun reliable capability
- Funding and attention shift as results disappoint or improve
- Benchmark gains do not automatically translate into safe deployment

KEY IDEA

Ask what works, for whom, and under what conditions.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Timeline activity

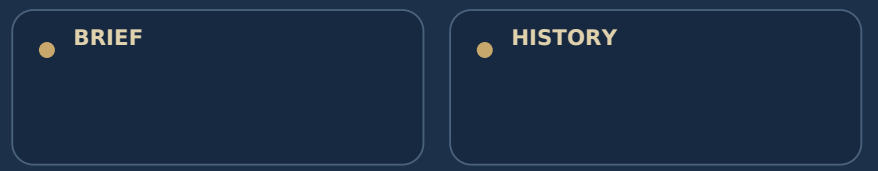
HOUR 03

- Place symbolic reasoning, expert systems, statistical learning, deep learning, and generative AI in order
- For each era, name one enabling factor
- Identify one limitation that persisted

KEY IDEA

Technology changes; the need for careful testing persists.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 3 synthesis

HOUR 03

- AI history is a sequence of representations, methods, and enabling conditions
- Each approach solves some problems and struggles with others
- Progress is neither linear nor guaranteed

KEY IDEA

Next: inspect the anatomy of an AI system.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

From problem to system

HOUR 04

- Specify a real user need and decision context
- Define what the system may and may not do
- Choose measurable outcomes and acceptable risks

KEY IDEA

A well-framed problem is half the engineering work.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Data: observations with a purpose

HOUR 04

- Collect only data relevant to the task
- Record provenance, permissions, and collection conditions
- Check missing values, imbalance, errors, and sensitive attributes

KEY IDEA

Data quality is contextual, not just numerical cleanliness.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Representations and features

HOUR 04

- Raw signals may need transformation before a method can use them
- Features summarize patterns relevant to a task
- Learned representations are produced by training rather than hand-designed rules

KEY IDEA

Representation choices shape what a model can notice.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Algorithm, model, and parameters

HOUR 04

- An algorithm is a procedure for learning or solving
- A model is the resulting structured predictor or decision process
- Parameters encode what training has fitted

KEY IDEA

Keep the learning procedure distinct from the learned artifact.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Training and inference

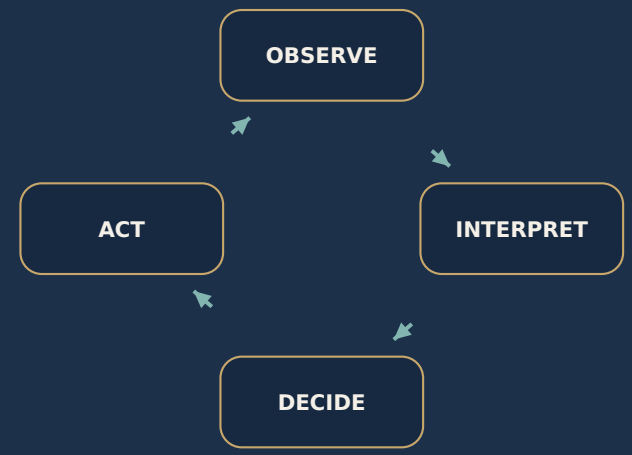
HOUR 04

- Training adjusts model parameters using examples and an objective
- Validation supports model and setting choices
- Inference applies the trained model to new inputs

KEY IDEA

Training learns; inference uses what was learned.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Evaluation beyond one accuracy number

HOUR 04

- Choose metrics tied to the actual decision
- Test on data not used to fit the model
- Examine subgroup performance, error types, and uncertainty

KEY IDEA

A metric is useful only when its meaning is understood.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Deployment is part of the system

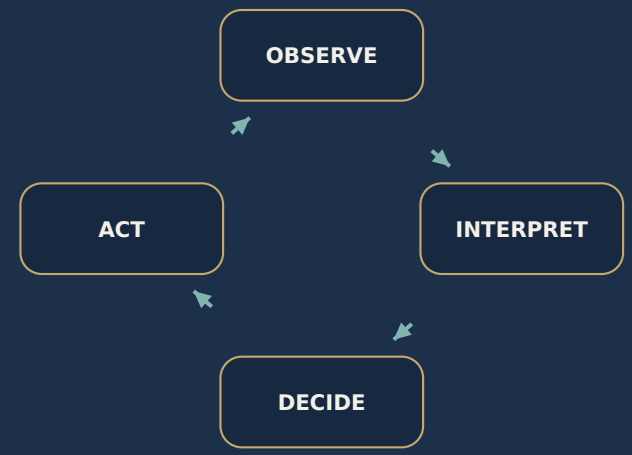
HOUR 04

- Integrate the model with interfaces and workflows
- Monitor changing inputs and outcomes
- Plan for versioning, rollback, incident response, and human escalation

KEY IDEA

A model is not a product until its surrounding system works.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Data leakage and misleading tests

HOUR 04

- Leakage lets information from the answer or future enter training
- Repeated tuning on a test set weakens the test’s independence
- Distribution shift changes how inputs differ after deployment

KEY IDEA

A clean evaluation asks a genuinely new question.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Walkthrough: plant-health classifier

HOUR 04

- Capture leaf images under varied lighting and backgrounds
- Label disease categories with expert review
- Train and validate; inspect false negatives by class
- Deploy with confidence thresholds and expert referral

KEY IDEA

The operating workflow matters as much as the classifier.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 4 synthesis

HOUR 04

- AI systems combine data, methods, people, and infrastructure
- Evaluation must reflect context and consequences
- Deployment requires monitoring and a plan for failure

KEY IDEA

Next: compare major AI approaches.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

A family of approaches

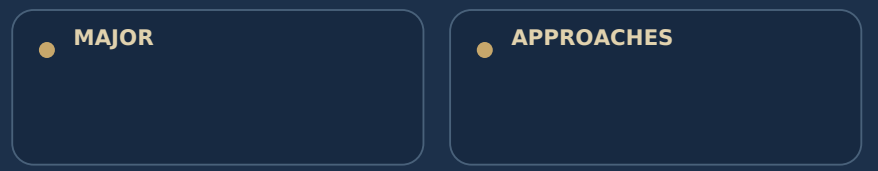
HOUR 05

- Rule-based reasoning uses explicit conditions
- Search explores candidate states or actions
- Machine learning estimates patterns from examples
- Hybrid systems combine methods

KEY IDEA

Choose by task constraints, not trendiness.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Rules and logic

HOUR 05

- Rules can encode policies or expert knowledge directly
- Logical inference can make the reasoning path explicit
- Exceptions and uncertainty can make rule sets hard to maintain

KEY IDEA

Rules work best when conditions are stable and expressible.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Search and planning

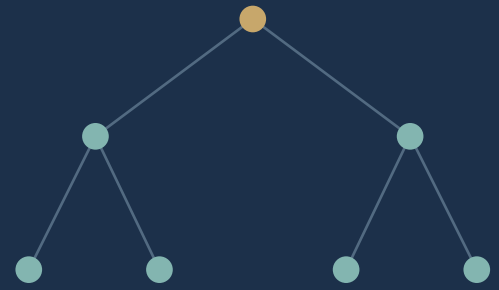
HOUR 05

- Define an initial state, available actions, and goal test
- Explore possible sequences of actions
- Use costs or heuristics to manage the search

KEY IDEA

Search is powerful when states and transitions are sufficiently modeled.

VISUAL MODEL



candidate states → goal

Conceptual illustration · not to scale

Knowledge representation

HOUR 05

- Represent entities, properties, relations, or constraints
- Use ontologies or graphs to organize domain concepts
- Reason over explicit relationships

KEY IDEA

Representation quality limits the conclusions that follow.

VISUAL MODEL



signals become relationships

Conceptual illustration · not to scale

Supervised learning

HOUR 05

- Learn a mapping from inputs to labeled examples
- Classification predicts categories; regression predicts numeric values
- Training data should represent intended use conditions

KEY IDEA

Labels encode human choices and may carry human error.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Unsupervised and self-supervised learning

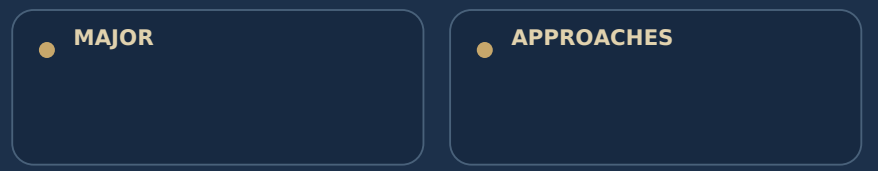
HOUR 05

- Find structure without task labels, such as clusters
- Learn useful representations from relationships within data
- Results require interpretation and task-specific validation

KEY IDEA

Discovered structure is not automatically meaningful.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Reinforcement learning

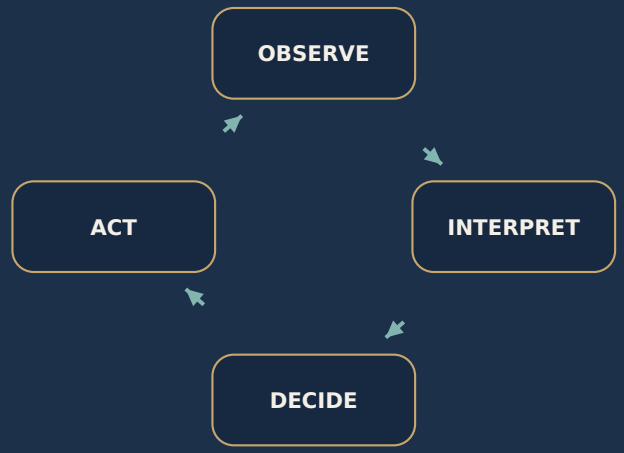
HOUR 05

- An agent acts and receives reward signals
- Learning balances short-term outcomes and future returns
- Reward design can create unintended behavior

KEY IDEA

The reward is a proxy for goals, not the goal itself.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Generative methods and hybrids

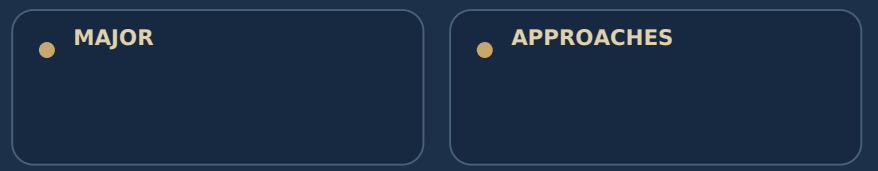
HOUR 05

- Generate new content conditioned on prompts or context
- Combine retrieval, rules, and models for grounded behavior
- Apply filters, provenance, and review where consequences matter

KEY IDEA

Generation expands capability but needs verification.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Approach-selection workshop

HOUR 05

- Specify task, data availability, and cost of errors
- Select a candidate method and explain why
- Name one alternative and one test that could change your choice

KEY IDEA

Method selection is an argument supported by evidence.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 5 synthesis

HOUR 05

- Rules, search, learning, and generation are complementary tools
- Each approach has assumptions and failure modes
- Hybrid architectures can allocate different tasks to suitable components

KEY IDEA

Next: connect AI to embodied systems and society.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

From digital model to physical action

HOUR 06

- A robot senses a changing physical world
- Software turns signals into estimates and choices
- Actuators turn decisions into forces and motion

KEY IDEA

Physical AI must account for time, uncertainty, and safety.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

The sense–think–act loop

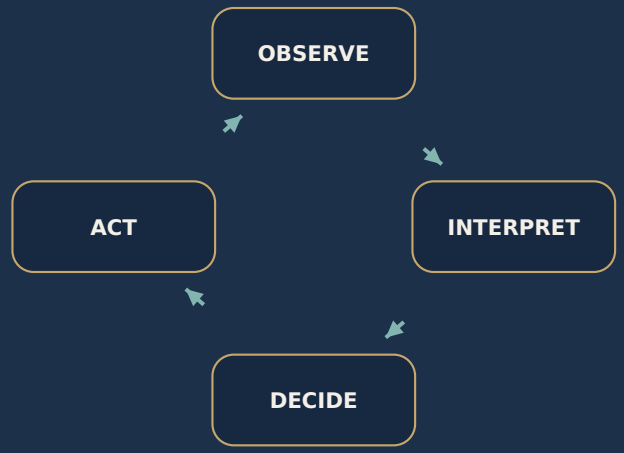
HOUR 06

- Sense: camera, range sensor, encoder, or touch
- Estimate: infer objects, position, and state
- Plan: choose a feasible action
- Act: move and measure the result

KEY IDEA

The loop repeats as the world changes.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Perception and sensor fusion

HOUR 06

- Sensors provide different views of the world
- Calibration relates measurements to a common frame
- Fusion combines evidence while preserving uncertainty

KEY IDEA

More sensors help only when their limits are understood.

VISUAL MODEL



signals become relationships

Conceptual illustration · not to scale

Planning and control are distinct

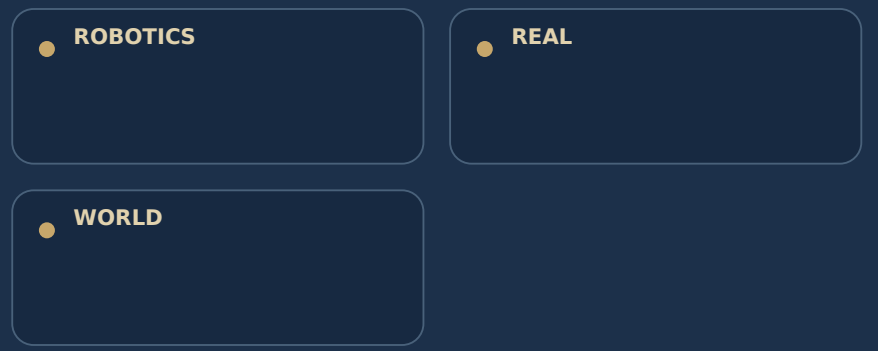
HOUR 06

- Planning proposes a goal-directed sequence
- Control computes commands that track the plan
- Feedback corrects deviations caused by disturbance or model error

KEY IDEA

Plans describe intent; controllers manage execution.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Examples across sectors

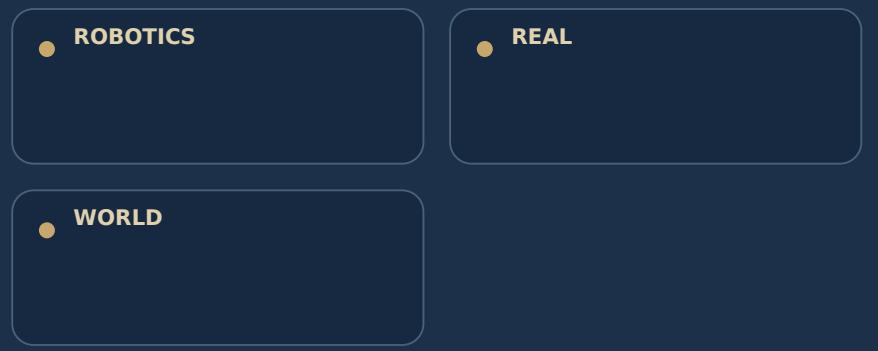
HOUR 06

- Agriculture: monitor crops and support precision tasks
- Healthcare: assist clinicians or support rehabilitation
- Industry: inspect, move, and collaborate on production lines
- Logistics: perceive, route, and handle goods

KEY IDEA

Applications should start with an actual user and context.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Human oversight and autonomy

HOUR 06

- Autonomy is a design choice on a continuum
- Humans may supervise, approve, intervene, or take over
- Interface design determines whether oversight is meaningful

KEY IDEA

A nominal human-in-the-loop is not enough; the person needs time and information.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Reliability in the physical world

HOUR 06

- Lighting, surfaces, wear, and obstacles alter sensor data
- Latency and hardware faults affect control
- Safe behavior may require conservative fallback modes

KEY IDEA

Test in conditions that resemble deployment.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Case: indoor delivery robot

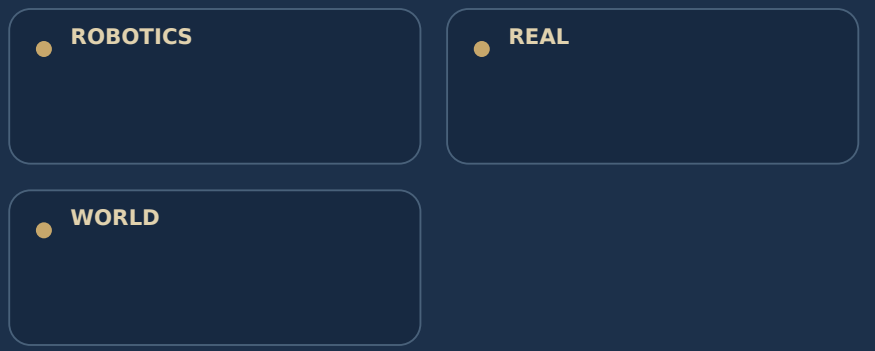
HOUR 06

- Map corridors and identify restricted areas
- Estimate location from sensors and map
- Plan a route around people and temporary obstacles
- Stop or request help when confidence or safety limits are crossed

KEY IDEA

A safe system knows when not to proceed.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Design a robot application

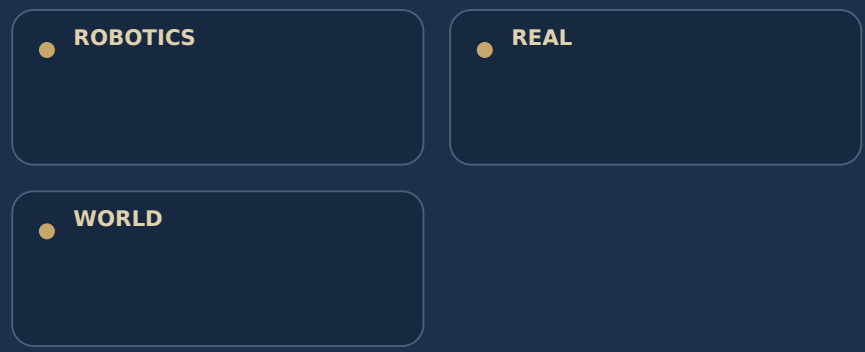
HOUR 06

- Choose a setting and a user need
- Sketch sensors, state estimate, planner, controller, and human interface
- Mark one hazard and a mitigation

KEY IDEA

Good robotics begins with task and hazard analysis.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 6 synthesis

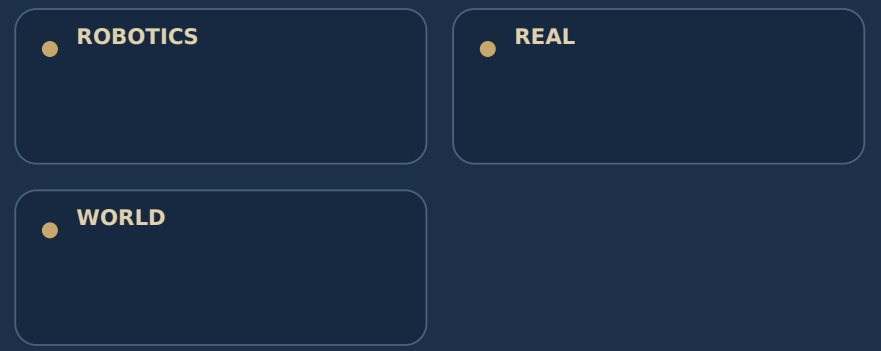
HOUR 06

- Robotics couples AI with sensing, mechanics, and control
- Uncertainty propagates from observation to action
- Human context and safe fallback are part of the design

KEY IDEA

Next: practice responsibility, evaluation, and critical reflection.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Capability and consequence

HOUR 07

- An accurate prediction can still be misused
- A low-probability error may have severe consequences
- Assess who benefits, who bears risk, and who can appeal

KEY IDEA

Technical performance is one part of responsible deployment.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Bias and representation

HOUR 07

- Historical data may reflect unequal access or treatment
- Sampling can omit communities or unusual conditions
- Labels and categories embed assumptions

KEY IDEA

Audit the full pipeline, not just the model output.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Privacy and data stewardship

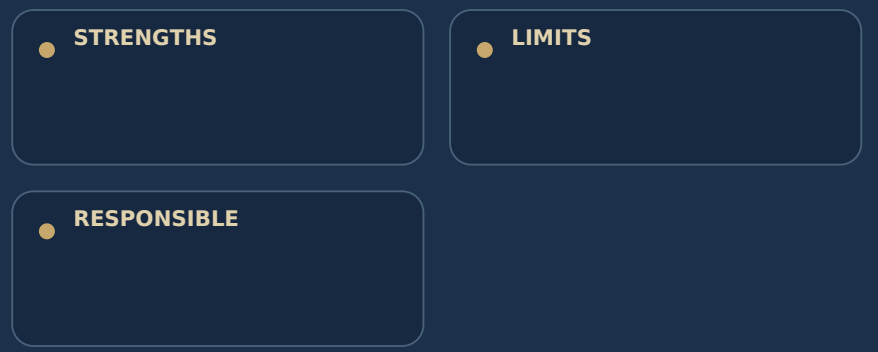
HOUR 07

- Collect only what is necessary
- Protect sensitive information through access controls and retention limits
- Explain data use and respect applicable consent and law

KEY IDEA

Privacy is a design requirement, not an afterthought.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Explainability and transparency

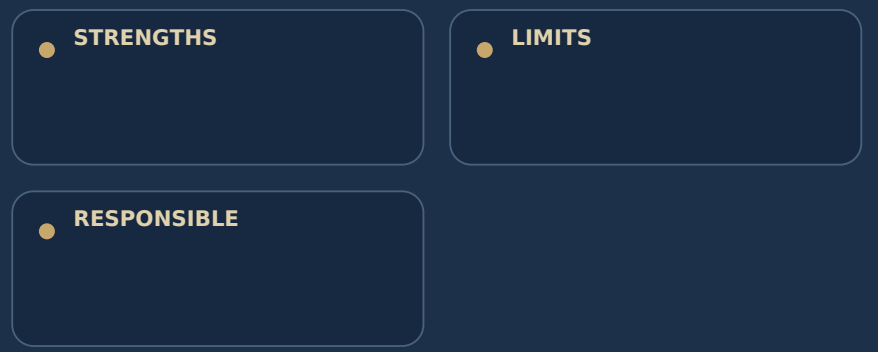
HOUR 07

- Different users need different explanations
- A feature attribution is not a complete causal explanation
- Communicate limitations, intended use, and confidence honestly

KEY IDEA

Explanations should support a real decision.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Uncertainty and calibration

HOUR 07

- A model can be uncertain even when it returns a confident-looking answer
- Calibration compares stated confidence with observed correctness
- Thresholds should reflect the cost of errors

KEY IDEA

Uncertainty can guide escalation and abstention.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Safety and robustness

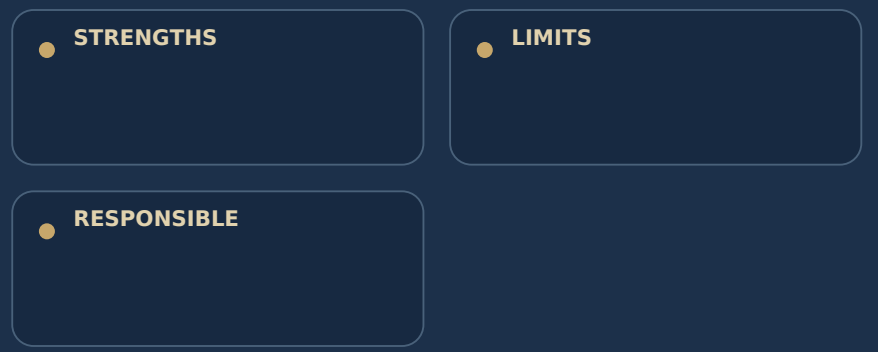
HOUR 07

- Test normal, edge, and adversarial conditions
- Use safeguards, constraints, monitoring, and fail-safe modes
- Document hazards and incident response

KEY IDEA

Safety is built through layers, not a single feature.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Human accountability

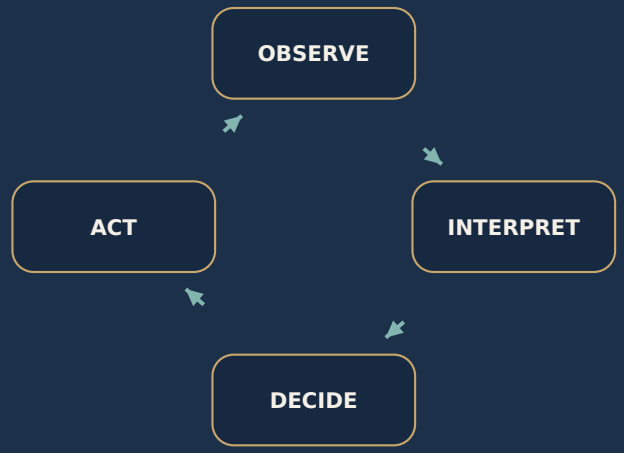
HOUR 07

- Assign a responsible owner for decisions and updates
- Ensure people can contest or correct consequential outputs
- Keep an audit trail proportionate to risk

KEY IDEA

Automation does not erase institutional responsibility.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Case: an automated admissions aid

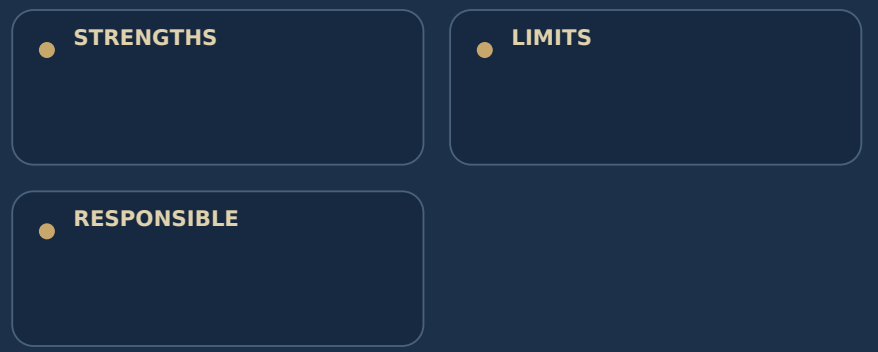
HOUR 07

- Potential benefit: organize large application queues
- Risks: proxy discrimination, inaccessible data, opaque rankings
- Controls: human review, validity testing, appeal route, monitoring

KEY IDEA

A high-stakes tool needs unusually strong evidence and governance.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Responsible-AI review exercise

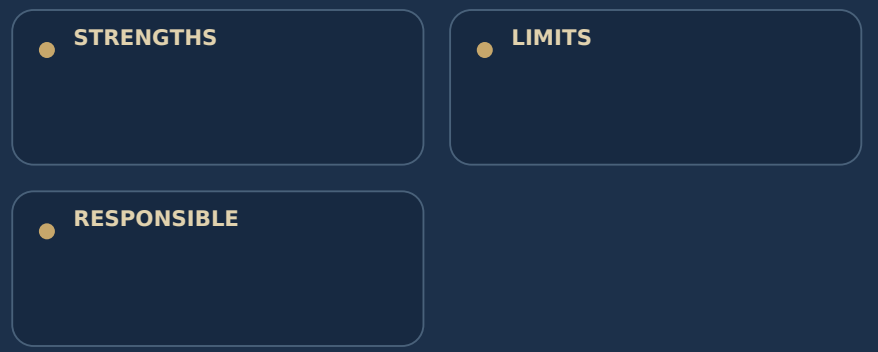
HOUR 07

- Who is affected and what is the decision?
- What evidence supports validity for this population?
- What happens when the system is wrong?
- Who can explain, contest, and repair harm?

KEY IDEA

If these questions lack answers, deployment is premature.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 7 synthesis

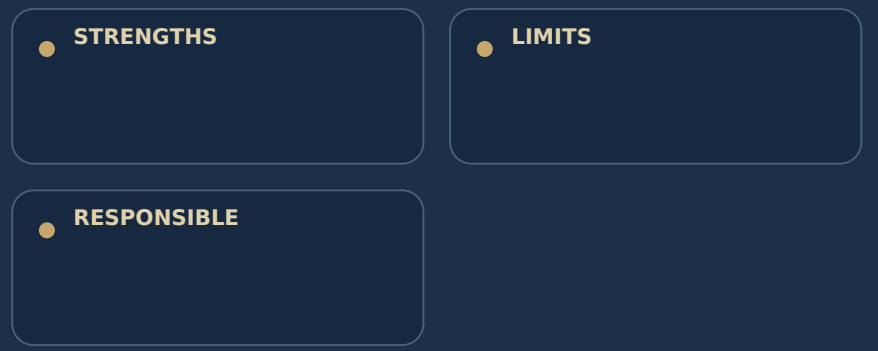
HOUR 07

- Fairness, privacy, safety, and accountability are socio-technical
- Model performance must be interpreted in context
- Responsible practice asks who is affected and how harm is addressed

KEY IDEA

Next: apply the concepts in a guided design challenge.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

The design challenge

HOUR 08

- Scenario: a greenhouse wants early warnings of crop stress
- Goal: help staff prioritize inspection, not replace their judgment
- Constraint: variable lighting, limited connectivity, and seasonal change

KEY IDEA

Design for a useful decision, not an impressive demo.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Step 1 — Frame the problem

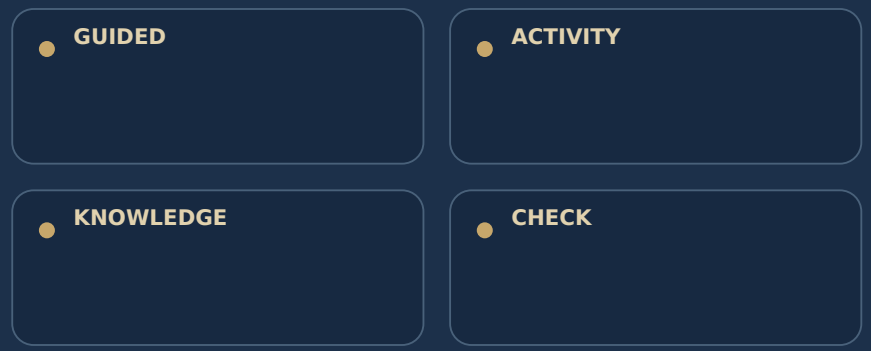
HOUR 08

- Identify the user and decision
- Define the intended benefit and what is out of scope
- Specify success and unacceptable failure

KEY IDEA

A narrow, testable scope is a strength.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Step 2 — Map inputs and outputs

HOUR 08

- Possible inputs: images, temperature, humidity, staff observations
- Output: prioritized alert with supporting evidence
- Human action: inspect, confirm, or dismiss

KEY IDEA

Trace every output to a real user action.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Step 3 — Choose an approach

HOUR 08

- Rule thresholds may flag known conditions
- A vision model may classify visible symptoms
- A hybrid can combine sensor rules and image evidence

KEY IDEA

Select a baseline before adding complexity.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Step 4 — Plan the data

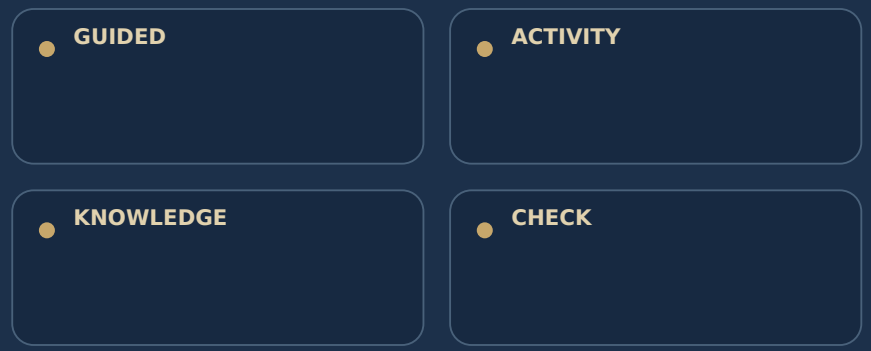
HOUR 08

- Sample crops, growth stages, weather, and lighting variation
- Obtain expert labels and record uncertainty
- Separate training and test data by plot or time to reduce leakage

KEY IDEA

The test should represent future conditions.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Step 5 — Define evaluation

HOUR 08

- Measure missed stress cases as well as false alarms
- Report performance by crop and condition
- Track time-to-inspection and whether staff find alerts useful

KEY IDEA

Evaluate the decision workflow, not only the model.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Step 6 — Add safeguards

HOUR 08

- Show uncertainty and reasons for alerts
- Allow staff to override and record corrections
- Escalate low-confidence cases; monitor seasonal drift

KEY IDEA

Safeguards make the system safer and more useful.

VISUAL MODEL



A system is a chain of choices and evidence.

Conceptual illustration · not to scale

Knowledge check

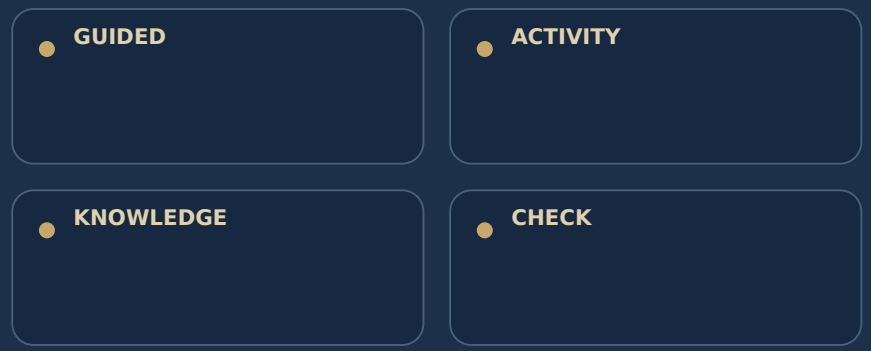
HOUR 08

- What distinguishes training from inference?
- Why is a held-out test set important?
- Name one source of data bias
- Why should a robot or AI system sometimes abstain?

KEY IDEA

Pause and answer before viewing the next slide.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Answer key & discussion

HOUR 08

- Training fits parameters; inference applies the fitted model
- Held-out data estimates generalization to unseen examples
- Sampling, labels, or historical decisions can introduce bias
- Abstention can reduce harm when evidence is insufficient

KEY IDEA

Explain the reasoning, not only the answer.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Hour 8 synthesis

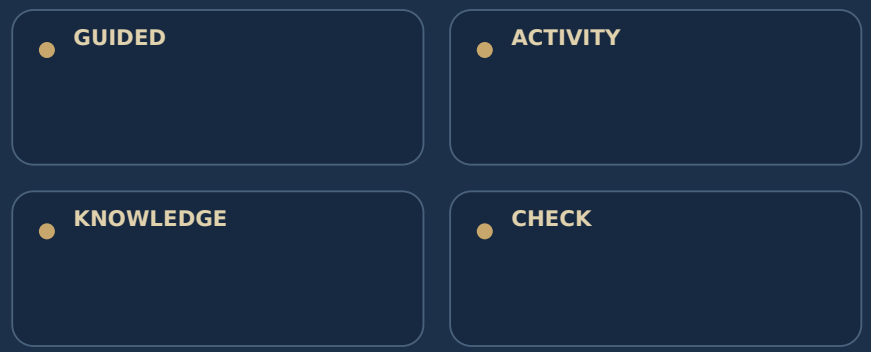
HOUR 08

- Frame the user decision before selecting a model
- Evaluation and safeguards belong in the initial design
- Human feedback can improve both system understanding and workflow

KEY IDEA

Next: consolidate the course and plan your next steps.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

The AI system in one picture

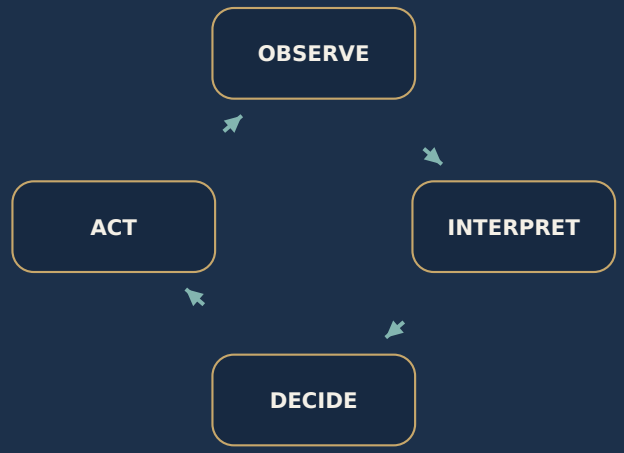
HOUR 09

- Problem context defines the task and acceptable outcomes
- Data and representations make the task computational
- Methods produce predictions, plans, or generated content
- People deploy, evaluate, govern, and improve the system

KEY IDEA

The system includes its human and institutional setting.

VISUAL MODEL



feedback updates the next step

Conceptual illustration · not to scale

Core vocabulary, part one

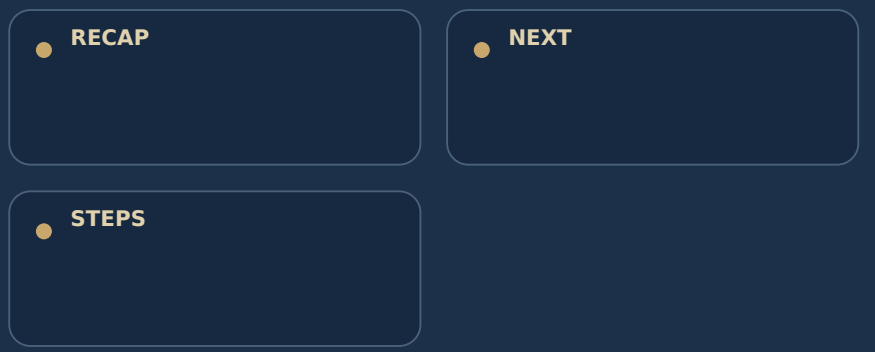
HOUR 09

- Agent: system that selects actions from observations
- Environment: world or context an agent interacts with
- Model: structured method or learned representation
- Training: fitting a model using data and an objective

KEY IDEA

Use terms consistently when discussing designs.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Core vocabulary, part two

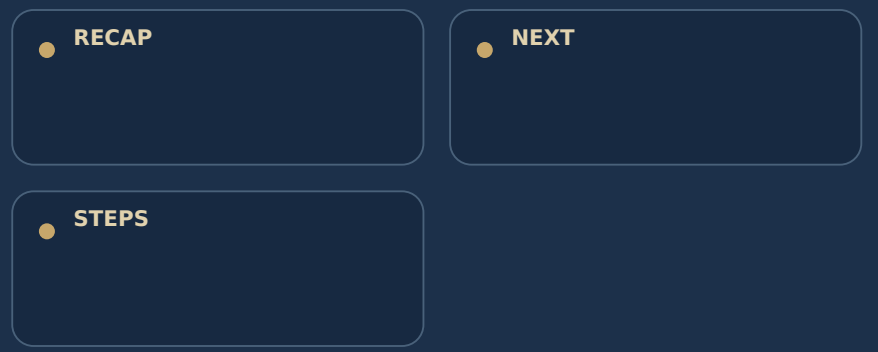
HOUR 09

- Inference: applying a trained model to inputs
- Feature: representation used to support a task
- Metric: quantitative measure of a defined outcome
- Distribution shift: changed input conditions after deployment

KEY IDEA

A metric needs context to become meaningful.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Five questions for any AI system

HOUR 09

- What task is being performed, and for whom?
- What data and assumptions shape the output?
- How was the system evaluated in relevant conditions?
- What happens when it fails or is uncertain?
- Who is accountable for outcomes and correction?

KEY IDEA

These questions travel across every AI specialization.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Common misconceptions to leave behind

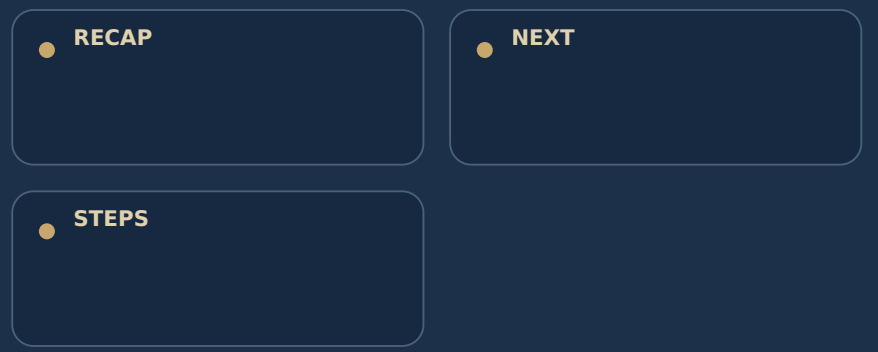
HOUR 09

- AI is not always learning or autonomous
- High accuracy does not guarantee fairness or safety
- Generated text can be fluent and still incorrect
- A model alone is not the complete deployed system

KEY IDEA

Healthy skepticism is compatible with technical ambition.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Your degree pathway

HOUR 09

- AI.102–106: robotics, learning, vision, language, and control foundations
- Year 2: perception, autonomy, reinforcement learning, and deep learning
- Years 3–4: integrated systems, specialized robotics, ethics, and project work

KEY IDEA

Build fundamentals steadily; later courses connect them.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Personal reflection

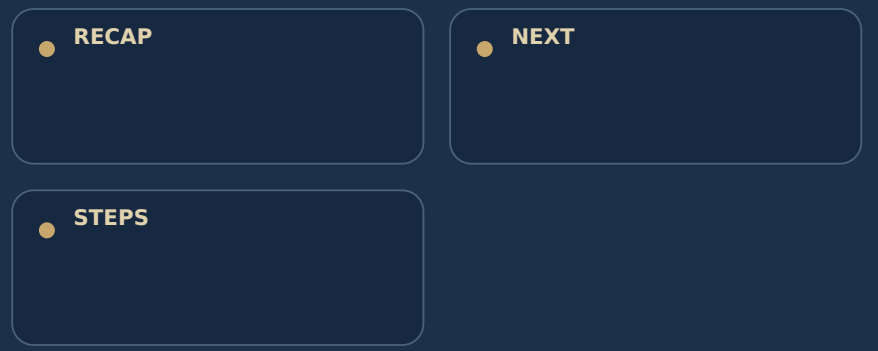
HOUR 09

- Which AI topic surprised you most?
- What assumption about AI changed during this orientation?
- What capability would you like to build responsibly?
- What concept needs more study?

KEY IDEA

Curiosity becomes expertise through deliberate practice.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Your first small investigation

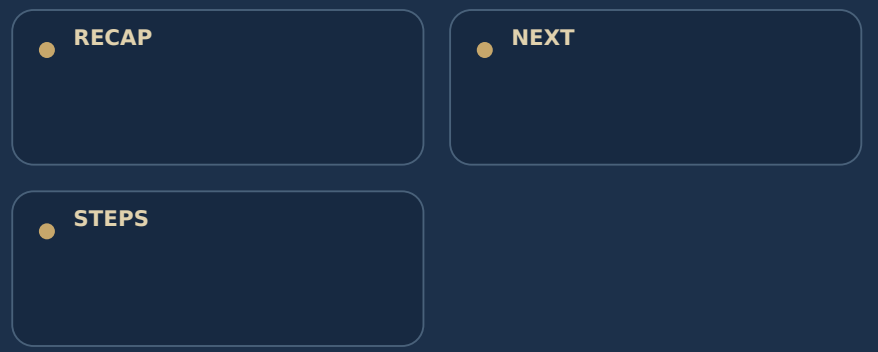
HOUR 09

- Choose an AI-enabled system you can observe
- Describe inputs, outputs, users, and consequences
- Find one claim about its performance and examine the evidence
- Suggest one improvement or safeguard

KEY IDEA

Practice critical analysis on real systems around you.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Closing knowledge check

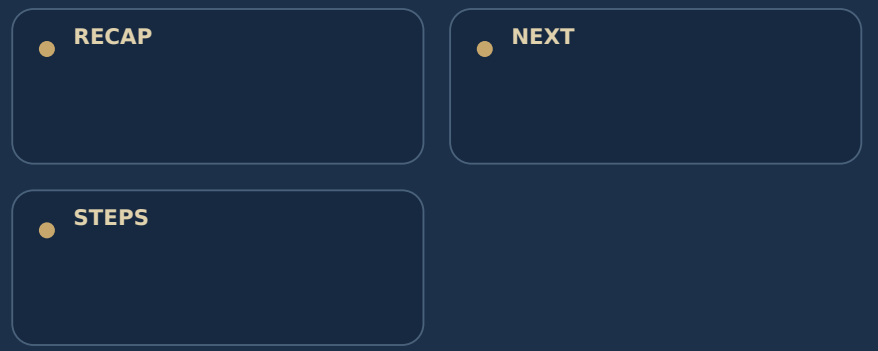
HOUR 09

- Explain AI in one careful sentence
- Draw a sense–think–act or data-to-deployment diagram
- Name one method, one evaluation concern, and one ethical risk
- Connect AI.101 to one later course

KEY IDEA

A strong foundation is the ability to ask better questions.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Thank you — begin with purpose

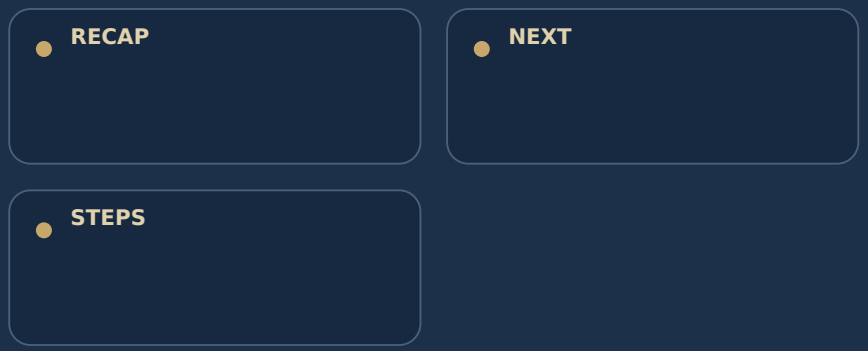
HOUR 09

- Observe systems carefully
- Build with evidence and humility
- Keep people and real-world context in view
- Carry responsible practice into every technical course

KEY IDEA

Intelligence in motion begins with thoughtful engineers.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix A · Guided study plan

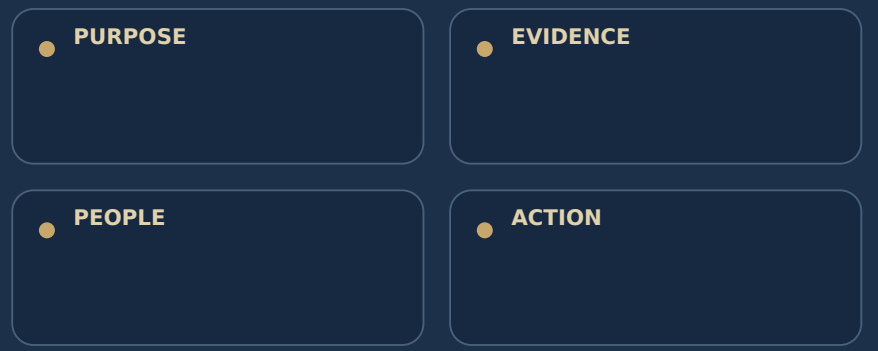
HOUR AP

- Before class: preview key terms and write one question
- During study: pause, sketch each system diagram, and retrieve concepts from memory
- After study: explain one idea to a peer without reading the slide
- Follow up: consult instructor-provided readings and laboratory materials

KEY IDEA

Use this deck alongside the university’s approved syllabus and readings.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix B · Glossary: foundations

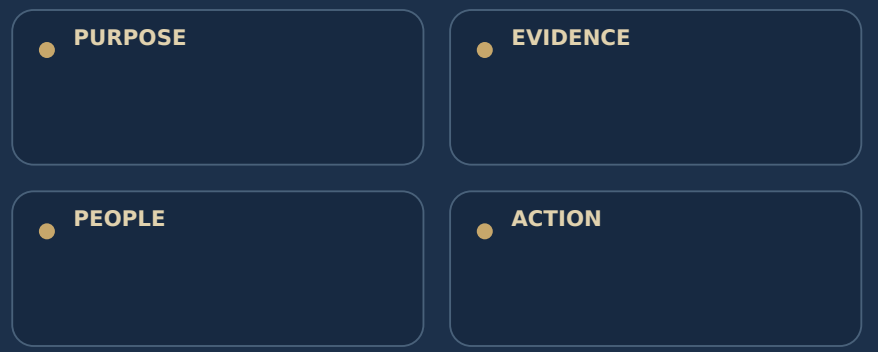
HOUR AP

- Agent: decision-making system interacting with an environment
- Environment: external context that provides observations and receives actions
- Algorithm: defined procedure for solving a problem or learning
- Model: structured predictor, representation, or decision process

KEY IDEA

Precise language improves both learning and engineering communication.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix C · Glossary: learning

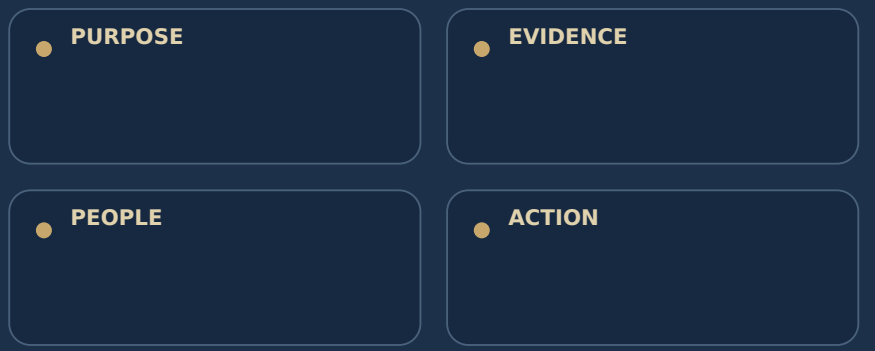
HOUR AP

- Dataset: organized examples used for development or evaluation
- Feature: input representation relevant to a task
- Label: target annotation associated with an example
- Generalization: performance on relevant unseen examples

KEY IDEA

Labels and datasets require documentation and scrutiny.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix D · Glossary: deployment

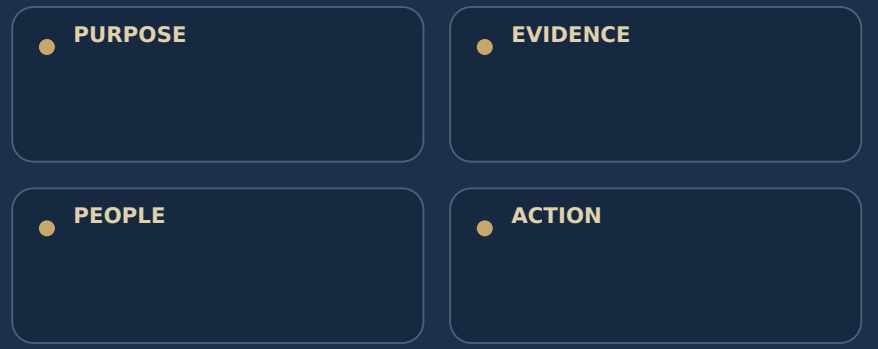
HOUR AP

- Inference: applying a model to new input
- Distribution shift: change in data conditions over time or context
- Calibration: agreement between confidence and observed correctness
- Fallback: safer alternative when system conditions are not met

KEY IDEA

Deployment introduces new conditions that must be monitored.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix E · Responsible-design checklist

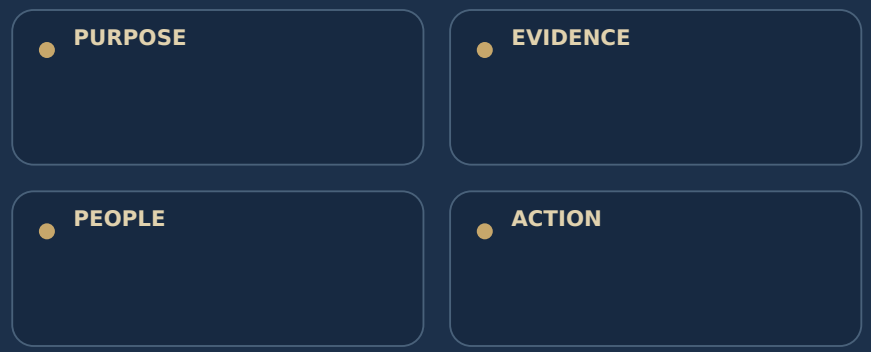
HOUR AP

- Purpose and affected people are clearly described
- Data rights, privacy, and provenance are addressed
- Testing matches the use context and includes error analysis
- Human review, correction, and incident handling are defined

KEY IDEA

Do not treat this checklist as legal or regulatory advice.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix F · Reflection prompts

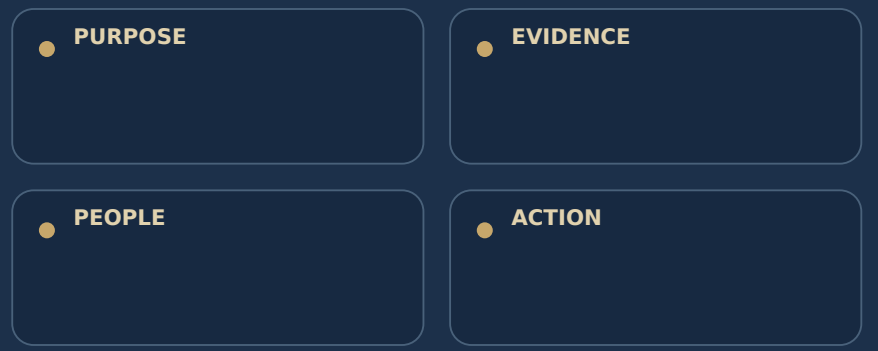
HOUR AP

- What is the simplest baseline that could meet the need?
- Which error would be most harmful, and to whom?
- What evidence would make you change the design?
- When should the system stop, defer, or ask a person?

KEY IDEA

Use reflection to uncover hidden assumptions.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Appendix G · Suggested next steps

HOUR AP

- Review course-specific syllabus, readings, and assessment guidance
- Strengthen programming, mathematics, and scientific communication
- Practice data visualization and experimental design
- Bring questions to instructors and lab sessions

KEY IDEA

This presentation is an orientation resource, not an official assessment specification.

VISUAL MODEL



Progress depends on context, evidence, and iteration.

Conceptual illustration · not to scale

Appendix H · Source and integrity note

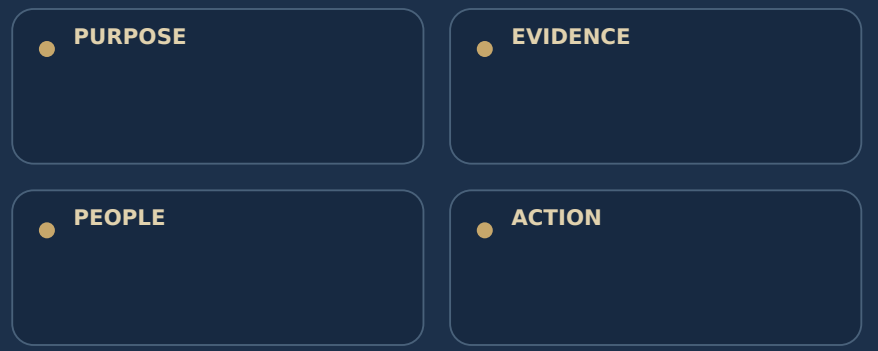
HOUR AP

- Explanations and diagrams in this deck are original instructional synthesis
- Examples are illustrative and should not be treated as product endorsements
- Verify current technical, legal, and institutional guidance before deployment
- Consult instructor-approved texts and references for formal study

KEY IDEA

Claims are intentionally framed as introductory, not exhaustive.

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale

Final slide · A principled beginning

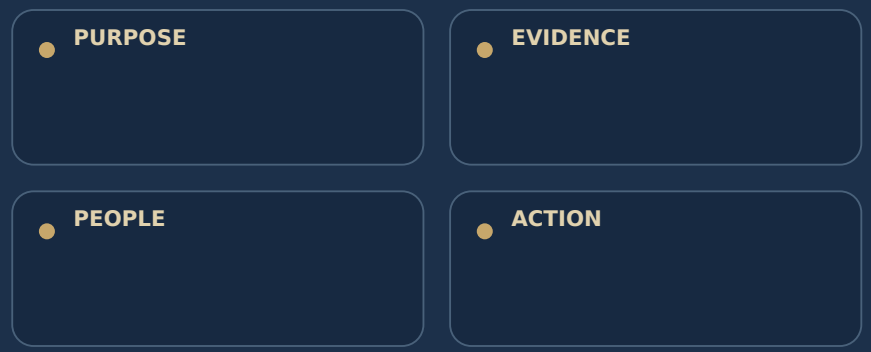
HOUR AP

- Understand the task
- Question the evidence
- Design for people and context
- Build systems that can be evaluated and corrected

KEY IDEA

TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

VISUAL MODEL



ASK → TEST → REFLECT

Conceptual illustration · not to scale